25-26 CS3003: IT Project Management

1. My courses
2. Introduction and Useful Information

Main content blocks

25-26 CS3003: IT Project Management

Main content blocks

Topic outline

- Collapse all

CS3003: IT Project Management

Welcome to the Moodle page for IT Project Management. The module aims to assist students in appreciating the scale and complexity of IT projects and provide real-world, highly-practical knowledge, best-practices and insights on how such projects can be effectively managed.

You may check the welcome video below for a quick introduction and a short description of the module.

Play Video

○

Announcements Forum

○

Q&A Forum

- **Weekly Content and Assigments**

Week 2 - 29/09/2025. Part 1. Project Management (PM) Fundamentals

Week 3 - 06/10/2025. Part 2. Project Pre-Initiation

Week 4 - 13/10/2025. Part 3. Project Initiation and Initial Planning

Week 5 - 20/10/2025. Part 4. Project Planning: Scope and Time

Assignment No1: Announced Thursday 23/10. Deadline: **10th November 2025, 14:00**

Week 6 - 27/10/2025 Part 4 (cont). Project Planning: Resources and Communications

Week 7 - 03/11/2025 Part 5. Project Risk Management

Assignment No2. Announced Tuesday 06/11. Deadline: **20th November 2025, 14:00**

Week 8 - 10/11/2025 Part 6. Project Controlling Processes: Configuration and Change Management

Week 9 - 17/11/2025 Part 7. Project Controlling Processes: Progress and Performance

Week 10 and Week 11 - 24th November and 01st December 2025 Part 8. Fundamentals of Team Management

Week 12 - 08th Dec. Part 9. Project Quality Management and revision

- **Introduction and Useful Information**

This section provides useful information regarding this module, namely a *detailed syllabus*, the intended *learning outcomes*, *reading resources* and an outline of *assessments* and *examination*. Moreover, in the context of assignments, you can find some useful resources on preparing presentations and on public speaking, an important skill for a Project Manager.

- **Week 2 - 29/09/2025. Part 1. Project Management (PM) Fundamentals**

Following two (motivational) videos that showcase the history, importance and some basic concepts (in a nutshell) of Project Management, the following topics will be covered in the first part of the module:

1. What a project is and what are a project's key characteristics
2. The typical lifecycle of a project
3. The project triangle and its significance
4. The distinguishing characteristics of technology projects in comparison to other domains (e.g. construction projects)

Moreover, we will discuss a case study of a typical real-world IT project which will subsequently be used in practical assignments and problem-solving exercises throughout the course.

ACTIONS for STUDENTS (Week 2 - 29/09/2025)

5. (**Highly recommended**) Study carefully the IT Project case study and consider the questions provided at the end of the document

- **Week 3 - 06/10/2025. Part 2. Project Pre-Initiation**

The second part of the module covers a number of concepts related to project tasks that are performed before a project is initiated (pre-initiation phase). Those are mainly the following:

Part 2.1: Business Case, Investment Appraisal, Estimations and Budgeting.

Part 2.2: IT Systems Acquisition Models and Procurement Management

- **Week 4 - 13/10/2025. Part 3. Project Initiation and Initial Planning**

The third part of the module covers a number of concepts and discusses the project tasks that are performed during the initiation and initial planning of a project. Those are mainly the following:

  - The notion of project stakeholders and the task of stakeholder analysis
  - The notion of project governance and the typical roles in the governance stucture of a project
  - The Project Charter

- **Week 5 - 20/10/2025. Part 4. Project Planning: Scope and Time**

Part 4 discusses concepts, tasks and pertinent technques and tools that are performed during the Planning phase of a project. In week No5, those are the following:

  - Requirements elicitation/definition (Project Scope)
  - The concept and technique of Work Breakdown Structure (WBS) (Project Scope)
  - Project Schedule Overview/Steps
  - Network Diagrams and Critical Path Analysis, the Project Gantt Chart and the Project Milestones

- **Assignment No 1 (25%)**

Due: Monday 10th November 2025, 14:00 a.m.

Please review the detailed assignment spec. for details. The spec. will be discussed in class on Thursday 23/10.

The assignment is **individual** and will be submitted via Moodle/Turnitin. **Questions regarding the assignment are allowed until Thursday November 6th, either in class or in the Q&A forum. Questions after that date, or sent via e-mail/Teams will not be answered!**

Please submit a single .zip file

Extensions

This assignment **is subject to Department policy on extensions**. If you believe you require an extension please read the documentation carefully **here**.

Extenuating circumstances

If you submit an assessment and believe that the standard of your work was substantially affected by your current circumstance then you can apply for Extenuating Circumstances. Details on how to apply for this can be found **here**

It is essential you read the accompanying documentation on the above link carefully. Please note decisions on Extenuating Circumstances are made at the end of the academic year.

Late Submission

In the absence of acceptable extenuating cause, late submission of work will be penalised as follows:

1. for work submitted up to 24 hours late, the mark will be reduced by ten percentage marks;

2. for work submitted more than 24 hours late, the maximum mark will be zero.

Academic Misconduct

Academic Misconduct Coursework submissions are routinely checked for academic misconduct (working together, copying from sources, etc.). Penalties can range from a 10% deduction of the assignment mark, zero for the assignment or the case being referred to a Senior Vice-Principal to make a decision (for repeat offences). Note that is academic misconduct if you fail to correctly reference or acknowledge where you have used generative AI tools to help you produce your submission. If the use of generative AI tools is prohibited for this assignment, then any use of such tools constitutes academic misconduct. Further details can be found **here.**

Use of AI

The use of generative AI tools (e.g. Microsoft Copilot, ChatGPT, Grammarly, etc.) is **not** permitted for this assignment. More information regarding the use of generative AI in your studies can be found **here.**

- **CS3003_OT_ALL_Assignment1 File**
- **CS3003_OT_ALL_Assignment1Turnitin Assignment**
- **CS3003Assignment No1_ Sample Answer File**
- **CS3003_Assignment1_SampleSolutionCostAnalysis File**

- **Week 6 - 27/10/2025 Part 4 (cont). Project Planning: Resources and Communications**

This part of the module covers the following topics, that are also part of the project planning phase.

-Resources Planning: Introduction and some common Tools/Techniques

-Basics of Project Communications Planning and Communications Management

- **Formative Revision Quiz**

This formative quiz aims to evaluate your understanding of some of the concepts that were discussed in the module. It should help you to self-evaluate and to identify gaps or misunderstandings in your knowledge. It does not count towards your final grade. Please note that a similar style of questions may be used in the module's exam/assessment.

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Formative Revision Quiz

- **Module Interim Feedback Survey**

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CS3003 Interim Feedback Survey

We would like to collect mid-term feedback aiming to find out what is working well in the module, what is not and how we can improve. This is even more important than ever because of the college-wide changes and the flexible learning model introduced this year. Completing the feedback survey should not take you more than 15 minutes! **The survey is anonymous and should be completed by the end of Monday 15th November .** Thanks in advance for your participation.

- **Week 7 - 03/11/2025 Part 5. Project Risk Management**

Part 5 of the module discusses fundamental concepts and best-practices related to Project Risk Management. In particular:

- Risk-related definitions and the Risk Management Plan

- The Risk Management Process, including the Risk Register, Best-Practices and Responsibilities

- **Formative Revision Quiz (Week 04/11)**

This **formative quiz** aims to evaluate your understanding of **Project Communications Management** and **Project Risk Management**. It should help you to self-evaluate and to identify gaps or misunderstandings in your knowledge. It does not count towards your final grade.

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Formative Revision Quiz (Week 04/11)

- **Week 8 - 10/11/2025 Part 6. Project Controlling Processes: Configuration and Change Management**

In Part 6 we will discuss fundamental concepts related to the Project Controlling Process Group. In particular, we will focus on:

- Configuration Management, including the differences between Software Configuration Management and Project Configuration Management.
- Change Management

- **Assignment No2 (25%)**

Due: Thursday 20th November 2025, 14:00

Please review the detailed assignment spec. for details. The spec. will be discussed in class on Thursday 06/11

The assignment is **individual** and will be submitted via Moodle/Turnitin. **Questions regarding the assignment are allowed until Friday November 14th, either in class or in the Q&A forum. Questions after that date, or sent via e-mail/Teams will not be answered!**

Please submit a single .pdf file (not a .zip file)

Extensions

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Senior Vice-Principal to make a decision (for repeat offences). Note that is academic misconduct if you fail to correctly reference or acknowledge where you have used generative AI tools to help you produce your submission. If the use of generative AI tools is prohibited for this assignment, then any use of such tools constitutes academic misconduct. Further details can be found **here.**

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- **CS3003_OT_ALL_Assignment2 File**
- **CS3003_OT_ALL_Assignment2Turnitin Assignment**
- **generalFeedbackCS3003AssignmentNo2 File**

- **Week 9 - 17/11/2025 Part 7. Project Controlling Processes: Progress and Performance**

In Part 7 we will discuss the remaining fundamental concepts of the Project Controlling Process Group.

In particular, we will focus on:

-S-Curve Analysis, Milestones, Tracking Gantt Charts
-Earned Value Management
-Scrum/Agile: Velocity and Sprint/Release Burndown Charts

- **Week 10 and Week 11 - 24 November and 01st December 2025 Part 8. Fundamentals of Team Management**

Highlighted

In Part 8 we will discuss fundamental concepts related to Teams and Teams Management. In particular, we will focus on:

-Main characteristics of effective project teams and reasons that teams fail

-Tuckman Model of Stages of Team/Group development

-Leadership Styles

-Basics of Goal-Setting

-Time Management

-Meetings Management

-Conflict Management

- **Week 12 - 08th Dec. Part 9. Project Quality Management**

Part 9 outlines some basic concepts related to Project Quality Management and Quality Management in general.

- **(Week 12 - 08th Dec.) Part 10. Project Manager Skills, Profession and Code of Ethics**

We are going to provide an overview of the skills and traits that a PM should have, the outlook and job market for the profession and, also, outline the relevant code of ethics. The contents of this section are not examinable (but they are still important!).

- **Interesting Resources & Articles**

In the links below you may find a number of interesting resources (usually recent articles) to complement your knowledge on Software Projects and on Technology Project Management. Note that those are not required for the module's assessment. Still, they can be very beneficial. This list will be gradually updated.

1. What consultancies look for in graduate project managers. November 2023. **https://www.apm.org.uk/blog/what-consultancies-look-for-in-graduate-project-managers/**

2. "**Project Managers, Unlock the Power of Timeboxing**", by Marc Zao-Sanders, Harvard Business Review (October 2023)

3. **"The Project Economy has Arrived"**, by Antonio Nieto-Rodriguez, Harvard Business Review (November-December 2021).

4. **"Agility Hacks"**, by Amy C. Edmondson and Ranjay Gulati, Harvard Business Review (November-December 2021).

5. **"Architecting the Future of Software Engineering: A National Agenda for Software Engineering Research & Development"**, SEI (Software Engineering Institute), November 2021

You may also wish to check out my personal, educational-oriented, **blog** in which I occasionally write about software projects and products.

- **Sample Exam Papers**

Topic outline

- **Introduction and Useful Information**

Collapse all

This section provides useful information regarding this module, namely a *detailed syllabus*, the intended *learning outcomes*, *reading resources* and an outline of *assessments* and *examination*. Moreover, in the context of assignments, you can find some useful resources on preparing presentations and on public speaking, an important skill for a Project Manager.

- **Syllabus Outline**

  1. Project Management (PM) Fundamentals
  2. Business Case and Investment Appraisal, Common Cost elements in IT projects; Models of S/W acquisition/licensing. Basics of Procurement and Contracts Management.
  3. Project Organization (e.g. Project Governance Structure; Stakeholders Analysis; Role of Project Manager)
  4. Basic Processes of a Project. References to relevant industry standards, such as PMI, ISO 21500 and Prince2.
  5. Project Planning
  6. Project Monitoring, Control and Implementation Processes
  7. Fundamentals of Team Management
  8. Project Risk Management: fundamental concepts and terminology; the risk management process.
  9. Quality Aspects
  10. The IT Project Manager Skills, Profession and Code of Ethics: introduction and outline of relevant topics.

- **Learning Outcomes**

  - LO1 - Understand the fundamental concepts, terminology and cost elements of IT Projects.
  - LO2 - Apply basic Project Planning techniques.
  - LO3 - Explain the procedures needed to monitor, control and report upon an IT Project.

- LO4 - Demonstrate an understanding of the steps needed to build/maintain effective development teams.
- LO5 - Discuss and where appropriate apply the principles of Project Risk Management.
- LO6 - Explain the ways in which appropriate quality attributes of the products of an IT Project can be assessed and assured.

Skills

- S1 - Problem Solving and Critical Thinking.
- S2 - Written Communication/Argumentation.
- S3 - Presentations skills.

- **Reading Resources**

1. Schwalbe, Kathy. Information Technology Project Management. Course Technology, (8th Ed), 2015, ISBN: 1285452348.

2. Field, Mike and Keller, Laurie. Project Management. Open University, International Thomson Press, 1998, ISBN: 1861522746.

3. Hughes, Bob and Cotterell, Mike. Software Project Management. McGraw-Hill (5th Ed), 2009, ISBN: 0077122798.

4. Brooks, Frederick. The Mythical Man-month: Essays on Software Engineering. Addison Wesley (2nd Ed), 1995, ISBN: 0201835959.

5. DeMarco, Tom and Lister, Tim. Peopleware: Productive Projects and Teams. Addison Wesley (3rd Ed), 2013, ISBN: 0321934113.

Additional readings and handouts will be provided on specific topics, e.g. relevant material from the Prince2 Manual and the PMBOK Guide, as well as papers and web resources.

- **Assessment and Examination**

There will be two assignments in total. Both assignments are individual, are assessed and are mandatory. Moreover, they are inspired by a real project case. More specific information regarding assignments is as follows:

- Assignment 1 (25%): aims to improve understanding and ability to apply some of the Project Cost, Initiation and Planning techniques (LO1, LO2)
- Assignment 2 (25%): aims to improve understanding of monitoring and controlling an IT Project (LO3) and apply fundamental principles of Project Risk Management (LO5).

Moreover, assignments aim to further develop skills in problem solving, analysis, critical thinking (S1), argumentation/decision making (S2) and communication/presentation skills (S3).

The list below provides **provisional** deadlines.

You may find the official departmental deadlines in the **departmental website**. Please note that there may be changes to the dates for coursework assignments which will be timely communicated.

- Assignment 1 (25%): Monday 10th November 2025
- Assignment 2 (25%): Thursday 20th November 2025

- **Useful Material On Presentations and Public Speaking**

Please find below some useful links, from TEDx, on preparing and delivering presentations and public speaking.

This is non-assessed material, but it's useful in order to be better prepared for assigments, in which you will be asked to put together a presentation/report.

Please review the material in the order provided.

1. "How to avoid death by Power Point", David JP Philips.

Play Video

;

2. "The magical science of storytelling", David JP Philips.

Play Video

3. "The 7 secrets of the greatest speakers in history", Richard Greene.

Play Video

4. "10 tips on how to make slides that communicate your idea", TED.com. https://blog.ted.com/10-tips-for-better-slide-decks/

- **Useful Links**

Course-related Links

- The Project Management Institute (PMI): https://www.pmi.org/
- Association for Project Management (APM): https://www.apm.org.uk/
- Prince2: https://www.axelos.com/best-practice-solutions/prince2

University-related Links

- Disability & Dyslexia Services: https://intranet.royalholloway.ac.uk/students/help-support/disabilities-and-dyslexia/home.aspx

- **Week 2 - 29/09/2025. Part 1. Project Management (PM) Fundamentals**

Collapse all

Following two (motivational) videos that showcase the history, importance and some basic concepts (in a nutshell) of Project Management, the following topics will be covered in the first part of the module:

1. What a project is and what are a project's key characteristics
2. The typical lifecycle of a project
3. The project triangle and its significance
4. The distinguishing characteristics of technology projects in comparison to other domains (e.g. construction projects)

Moreover, we will discuss a case study of a typical real-world IT project which will subsequently be used in practical assignments and problem-solving exercises throughout the course.

ACTIONS for STUDENTS (Week 2 - 29/09/2025)

5. (**Highly recommended**) Study carefully the IT Project case study and consider the questions provided at the end of the document

- **Videos on the History and Importance of Project Management**

The Giza Pyramid Project (History of Project Management Book).

Play Video

The History of Project Management (a part of it), video created for the 40 years of the Association for Project Management (APM), the UK chartered body for the project profession.

Play Video

- **Micro-Lectures, Slides and Resources (PM Fundamentals)**

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Introduction to PM Part1 (Video)(12min:37secs)URL

- CS3003 Part 1. Project Management Fundamentals (Slides of Week 2) File

- CS3003_CaseStudy File

- Quiz (formative)

This quiz aims to evaluate your understanding of some of the concepts introduced in the first week. It should help you to self-evaluate and to identify gaps or misunderstandings in your knowledge.

- **Week 3 - 06/10/2025. Part 2. Project Pre-Initiation**

Collapse all

The second part of the module covers a number of concepts related to project tasks that are performed before a project is initiated (pre-initiation phase). Those are mainly the following:

Part 2.1: Business Case, Investment Appraisal, Estimations and Budgeting.

Part 2.2: IT Systems Acquisition Models and Procurement Management

  - **Slides**

    - CS3003_Part2_Project PreInitiation File

  - **Surgeries**

    - Surgery No1 File

    - Surgery No1 Solutions File

  - **Handouts**

    - Simple Example of a Business Case (from Schwalbe's book) File

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Main Types of Contracts (from Schwalbe's book) File

- **Week 4 - 13/10/2025. Part 3. Project Initiation and Initial Planning**

Collapse all

The third part of the module covers a number of concepts and discusses the project tasks that are performed during the initiation and initial planning of a project. Those are mainly the following:

  - The notion of project stakeholders and the task of stakeholder analysis
  - The notion of project governance and the typical roles in the governance stucture of a project
  - The Project Charter
  - **Slides and Resources**

▪

CS3003_Part3_Project Initiation and Initial Planning (Slides) File

Week 5 - 20/10/2025. Part 4. Project Planning: Scope and Time

Collapse all

Part 4 discusses concepts, tasks and pertinent technques and tools that are performed during the Planning phase of a project. In week No5, those are the following:

- Requirements elicitation/definition (Project Scope)
- The concept and technique of Work Breakdown Structure (WBS) (Project Scope)
- Project Schedule Overview/Steps
- Network Diagrams and Critical Path Analysis, the Project Gantt Chart and the Project Milestones
- **Slides and Resources**

  - CS3003_Part4_Project Planning_Scope (Slides) File
  - CS3003_Part4_Project Planning_Time (Slides) File
  - Project Planning (Scope and Requirements) (Video, 15min)URL

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CS3003 Network Diagram Example File

- **Surgeries**

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SurgeryNo2_ProjectPlanning_Time File

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CS3003_Surgery No2_Answer_ProjectLibreGanttChart File

-

SurgeryNo2_ProjectPlanning_Time_Answers File

- **Handouts**

-

Project Schedule Gantt Diagram (from Schwalbe's book) File

-

Project Schedule Precedence Diagram (from Schwalbe's book) File

- **Links to Videos**

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ServiceNow SPM Project Workspace IntroductionURL

-

ServiceNow SPM Project Workspace (Detailed Video)URL

-

Microsoft Project Demo (Overview and some details)URL

-

Monday.com (PM Overview)

- **Week 6 - 27/10/2025 Part 4 (cont). Project Planning: Resources and Communications**

Collapse all

This part of the module covers the following topics, that are also part of the project planning phase.

-Resources Planning: Introduction and some common Tools/Techniques

-Basics of Project Communications Planning and Communications Management

- **Slides and Resources**

CS3003_Part 4_Planning_Resources_and_Communications (Slides) File

- **Week 7 - 03/11/2025 Part 5. Project Risk Management**

Collapse all

Part 5 of the module discusses fundamental concepts and best-practices related to Project Risk Management. In particular:

- Risk-related definitions and the Risk Management Plan

- The Risk Management Process, including the Risk Register, Best-Practices and Responsibilities

- **Slides and Resources**

CS3003_Part5_Project Risk Management (Slides) File

- **Surgeries**

CS3003 Surgery No3 - Project Risk Management File

CS3003_SurgeryNo3_Project Risk Management_Answers File

- **Handouts**

Prince2 Risk Categories File

- **Week 8 - 10/11/2025 Part 6. Project Controlling Processes: Configuration and Change Management**

Collapse all

In Part 6 we will discuss fundamental concepts related to the Project Controlling Process Group. In particular, we will focus on:

- Configuration Management, including the differences between Software Configuration Management and Project Configuration Management.
- Change Management

- **Slides and Resources**

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CS3003_Part6_Configuration and Change Management (Slides) File

- **Week 9 - 17/11/2025 Part 7. Project Controlling Processes: Progress and Performance**

Collapse all

In Part 7 we will discuss the remaining fundamental concepts of the Project Controlling Process Group.

In particular, we will focus on:

-S-Curve Analysis, Milestones, Tracking Gantt Charts
-Earned Value Management
-Scrum/Agile: Velocity and Sprint/Release Burndown Charts

- **Slides and Resources**

  -

CS3003_Part7_Project_Controlling_Progress_Performance (Slides) File

- **Surgeries**

  -

CS3003 Surgery Questions File

- **Formative Quiz**

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Formative Quiz (Project Progress - Earned Value Analysis)

This formative quiz aims to evaluate your understanding of some of the concepts that were discussed in the Project Control/Progress & Performance part of the module. It should help you to self-evaluate and to identify gaps or misunderstandings in your knowledge. It does not count

towards your final grade. Please note that a similar style of questions may be used in the module's exam/assessment.

- **Week 10 and Week 11 - 24 November and 01st December 2025 Part 8. Fundamentals of Team Management**

Highlighted

Collapse all

In Part 8 we will discuss fundamental concepts related to Teams and Teams Management. In particular, we will focus on:

-Main characteristics of effective project teams and reasons that teams fail

-Tuckman Model of Stages of Team/Group development

-Leadership Styles

-Basics of Goal-Setting

-Time Management

-Meetings Management

-Conflict Management

  - **Slides and Resources**

Highlighted

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CS3003_Part8_Fundamentals of Team Management (Slides) File

- **Week 12 - 08th Dec. Part 9. Project Quality Management**

Collapse all

Part 9 outlines some basic concepts related to Project Quality Management and Quality Management in general.

  - **Slides and Resources**

    -

CS3003_Part9_Project_Quality_Management File

- **(Week 12 - 08th Dec.) Part 10. Project Manager Skills, Profession and Code of Ethics**

Collapse all

We are going to provide an overview of the skills and traits that a PM should have, the outlook and job market for the profession and, also, outline the relevant code of ethics. The contents of this section are not examinable (but they are still important!).

- **Slides and Resources**

  -

CS3003_Part10_PM Skills and Code of Ethics File